ניתוח יציבות ודיוק של סכימות שיכון אינוריאנטיות לתמורות

חיבור על מחקר

לשם מילוי חלקי של הדרישות לקבלת התואר
מגיסטר למדעים במתמטיקה

רווינה רווינה

הוגש לסנט הטכניון - מכון טכנולוגי לישראל
אלול התשפ"ג, חיפה, ספטמבר 2023

Analysis of Stability and Accuracy of Permutation Invariant Embedding Schemes

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Analysis of Stability and Accuracy of Permutation Invariant Embedding Schemes

Research Thesis

Submitted in partial fulfillment of the requirements for the degree of Master of Science in Mathematics

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Submitted to the Senate of the Technion—Israel Institute of Technology

Elul 5783, Haifa, September 2023

The Research Thesis Was Done In The Faculty Of Mathematics
Under The Supervision of Dr. Nadav Dym.

The author of this thesis states that the research, including the collection, processing, and presentation of data, addressing and comparing to previous research, etc., was done entirely in an honest way, as expected from scientific research that is conducted according to the ethical standards of the academic world. Also, reporting the research and its results in this thesis was done in an honest and complete manner, according to the same standards.

Publication: Theorem 6.0.1 in the thesis was included in the paper [1] as Theorem 5.1. The other results in the paper were not related to the thesis. The work in this thesis is solely based on the work of Mrs. Ravina and Dr. Dym, the other authors of the paper were not involved in it. This paper can be found in DOI: 10.48550/arXiv.2306.06529

The generous financial help of the Technion is gratefully acknowledged.

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1 Abstract

Permutation-invariant neural networks is a type of deep learning architecture that can recognize patterns in sets even if the inputs are in a different order. Many of these architectures have the separation property meaning that these architectures map distinct inputs to distinct outputs up to permutation. We call such networks separating permutation invariant neural networks (SPINNs).

To ensure the robustness (stability) of these neural architectures we want the neural network to be Bi-Lipschitz. For that, we analyzed the Bi-Lipschitz property of different types of permutation-invariant functions to see how well these functions can be used to create stable embeddings for SPINNs. We proved that the permutation-invariant embeddings obtained by the summation of smooth or piecewise smooth functions are not Bi-Lipschitz. Also, it is already known that embeddings that combine linear mappings and 1-dimensional sorting are injective and permutation invariant and that the Bi-Lipschitz constants can be characterized in terms of the singular values of the linear component of the embedding. We improved these results by proving injectivity with an embedding dimension lower by a factor of two than the previously known dimension and improved the estimates for the Bi-Lipschitz constants of these embedding schemes.

We performed some experiments to analyze the stability and accuracy of the sorting-based SPINNs. These experiments produced some interesting observations and new insights into the stability and accuracy of the model. On our synthetic dataset, the model achieved high stability and accuracy even when the embedding dimension was very low compared to the required embedding dimension proved in our theoretical results.

2 Abbreviations and Notations

- $\mathbb{N}$: *The set of natural numbers*
- $\{1, 2, \ldots, n\}$: *The set of natural numbers from 1 to n*
- $|X|$: *The number of elements in set X*
- $\mathbb{R}$: *The set of real numbers*
- $\mathbb{R}^d$: *Set of vectors of length d*
- $\mathbb{R}^{d\times n}$: *Set of all $d \times n$ matrices*
- $(a_{ij})_{n\times n}$: *$n \times n$ Matrix with element a_{ij}*
- S_n : *Symmetric group (set of all $n \times n$ permutation matrices)*
- C^∞ : *Set of infinite differentiable functions*
- $\max(S)$: *The maximum of set S*
- $\min(S)$: *The minimum of set S*
- $n!$: *Factorial of a natural number n*
- $[X]$: *Equivalence class of X*
- $\mathbb{R}^n/S_n$: *Set of all the equivalence classes*
- $\|X\|$: *Euclidean norm of vector X*
- $\|A\|_F$: *Frobenious norm of matrix A*
- $\langle X, Y\rangle$: *Frobenius inner product on matrices*
- $\sim$: *Equivalence relation*
- M, m : *Upper and lower Lipschitz constants*
- π : *Permutation element in S_n*
- $\sum$: *Summation operator*
- $Conv(S)$: *Convex hull of a set $S \subset \mathbb{R}^d$*
- $O(\lambda^n)$: *Of order λ^n*
- DS_n : *Set of doubly stochastic matrices*
- Ω : *Set notation*
- $J(p)$: *Jacobian of a function p*
- $\approx$: *Approx*
- $det(A)$: *Determinant of matrix A*
- ker : *Kernel of a map*

- $\downarrow$: *Sorting in increasing order*
- σ_d : *d-th singular value of matrix*
- $\sigma_{d,k}$: *Minimal* d^{th} *singular value of all possible* $k \times d$ *sub-matrices of a matrix*

3 Introduction

With the advancement in the field of artificial intelligence deep learning frameworks proved to be effective for solving different types of problems such as image classification, text generation, language translation, etc. In each type of task to be solved, these frameworks make use of the underlying structure of the input data. There are numerous problems in which the underlying data is in the form of a *set*. That is, where the order of the elements does not matter and the task is to find an algorithm that produces the right results irrespective of the order of elements in the input dataset.

Let $\mathbb{R}^{d\times n}$ be the space of all sets containing n number of points of dimension d and S_n be the set of all the permutations on the n symbols. We sometimes identify a permutation $\pi \in S_n$ with the corresponding permutation matrix P. Taking this into consideration let us define the following definition.

Definition 3.0.1 (X$\sim$ Y)**.** Let $X, Y \in \mathbb{R}^{d\times n}$ then $X \sim Y$ if for some $P \in S_n$, $X = YP$.

Definition 3.0.2 (Permutation Invariant Function)**.** A function $f : \mathbb{R}^{d\times n} \to \mathbb{R}^D$ is said to be permutation invariant if for every permutation $\pi \in S_n$ and $(X_1, \ldots, X_n) \in \mathbb{R}^{d\times n}$,

$$f(X_1, X_2, X_3, ...X_n) = f(X_{\pi(1)}, X_{\pi(2)}, X_{\pi(3)}, ...X_{\pi(n)})$$

Definition 3.0.3. Let $X, Y \in \mathbb{R}^{d\times n}$ and S_n be the group of all permutations on n symbols. We define a metric on $\mathbb{R}^{d\times n}/\sim$ as follows:

$$d(X, Y) = \min_{P\in S_n} \|X - YP\|_F$$

Where this norm is the frobenious norm.

Note: In this thesis the norm of vectors is always the $L2$ norm and the norm of matrices is always the Frobenious norm.

To solve problems defined on sets, permutation-invariant neural networks are gaining increasing popularity in deep learning. These networks are useful in tasks such as set-to-set or set-to-sequence learning where the input and output data are unordered, because such networks can recognize patterns in a set of inputs, even if the inputs are in different orders. Recent work on permutation invariant neural networks include[20, 14, 9, 10, 16, 2, 17].

One of the most well-known permutation invariant architectures is "DeepSets" proposed by Zaheer et al (2017)[20]. The structure of DeepSets architecture on $\mathbb{R}^{d\times n}$ is of the form:

$$\phi(X_1, ..., X_n) = \rho(\sum_i f(X_i))$$

where the maps ρ and f are typically chosen to be neural networks.

DeepSets apply the same neural network function to each element of a set and then aggregate the output using a permutation-invariant function, such as the sum or the mean.

For example, let's say we want to classify a set of 2D points into two categories, "circle" and "square." The DeepSets architecture takes each 2D point in the set as input, uses a neural network to each point to generate a feature vector, and then applies a permutation-invariant function such as sum or mean to the feature vectors to produce a single fixed-size output vector. The output vector is then fed into a final classification layer to classify the set as a "circle" or "square".

Separability: The DeepSets architecture can approximate any continuous permutation invariant function on $\mathbb{R}^{d\times n}$, providing that the function $\sum_i f(X_i)$ is *separating*.

Definition 3.0.4 (Separation)**.** A permutation invariant function f is separating on $\mathbb{R}^{d\times n}$ if it is injective on $\mathbb{R}^{d\times n}/S_n$. Here we identify f with the induced function $\hat{f}$ on $\mathbb{R}^{d\times n}/S_n$ defined by $\hat{f}([X]) = f(X)$.

Lots of work has been done on the injectivity of the neural networks such as in[13, 8, 2, 20, 17, 12, 7].

In the context of the mapping $f : \mathbb{R}^{d\times n} \to \mathbb{R}^D$, we will refer to D as the embedding dimension. To ensure the injectivity of function f, it is crucial to optimize the required minimum embedding dimension to achieve this injectivity. The choice of the embedding dimension significantly influences the computational cost of the model. The higher the embedding dimension, the greater the computational cost of the model. Wagstaff et al. [17] proved that the necessary embedding dimension for the map $f : \mathbb{R}^n \to \mathbb{R}^D$ to maintain injectivity is at least n, which corresponds to the input set's size. Subsequent research on the embedding dimension of permutation invariant embedding schemes has been conducted by Balan et al [2] and Dym et al [7] where they provided the bounds on the minimum required embedding dimension to achieve the injectivity of f.

Stability: Performance of these networks for a given task (classification or regression) can be sensitive to input perturbations. Even a minor perturbation in the input can result in a substantial accuracy reduction of the model. Recent works by Rosca et al., Cahill et al. and others have investigated the stability of machine learning models through the study of their Bilipschitz continuity [15, 3].

Definition 3.0.5 (Bi-Lipschitz)**.** A map $f : \mathbb{R}^{d\times n} \to \mathbb{R}^D$, is said to be Bi-Lipschitz if there are constants m *and* M such that for all $X, Y \in \mathbb{R}^{d\times n}$,

$$m \cdot d(X, Y) \leq \|f(X) - f(Y)\|_2 \leq M \cdot d(X, Y)$$

where d is defined in Definition 3.0.3.

To embed $\mathbb{R}^{d\times n}/\sim$ into the embedding space $\mathbb{R}^D$, two permutation invariant embedding schemes are discussed by Balan et al [2] i.e.

- Algebraic Embedding obtained by power sum function and
- Sorting-based embedding scheme.

We will discuss both embeddings one by one.

Algebraic embeddings given by the power sum function defined below in section5 are discussed by Balan et al in [2]. They said that algebraic embeddings obtained by the power sum function on $\mathbb{R}^{d\times n}$ are not Lipschitz. In this work, we give full proof of this fact by proving that the power sum function for $d = 1$ is not Bi-Lipschitz and then we generalize this result for $d \geq 1$ by showing that all embeddings obtained by smooth or summation of piecewise smooth functions are not Bi-Lipschitz. Therefore any embedding obtained by smooth or summation of piecewise smooth functions can't be used to obtain a robust permutation invariant deep learning model.

Next, they discussed sorting-based permutation invariant embedding schemes[2]. Let us begin with one-dimensional sorting as follows

Definition 3.0.6 (Sorting)**.** Let $X \in \mathbb{R}^n$, then the sorting map $\downarrow$: $\mathbb{R}^n \to \mathbb{R}^n$ sorts $X = (x_1, x_2, \ldots, x_n)$ in increasing order. That is $\downarrow (x_1, x_2, \ldots, x_n) = (x_{p(1)}, x_{p(2)}, \ldots, x_{p(n)})$ such that $x_{p(1)} \leq x_{p(2)} \leq \ldots, \leq x_{p(n)}$, where $p \in S_n$ is a permutation .

When $d = 1$, it is trivial to see that sorting is a separating map. Moreover, it is an isometry as stated in [17], we will prove this result in Proposition 7.1.1. Being an isometry it is Bi-Lipschitz.

For $d \geq 1$ we define the row-wise sorting map on $\mathbb{R}^{d\times n}$ which we will use in our further discussion of sorting based permutation invariant embedding schemes.

Definition 3.0.7 (Row-Sorting $\downarrow$)**.** Let $X \in \mathbb{R}^{d\times n}$ then the map row sorting sorts each row of matrix X in increasing order independently. That is for each row $X^* = (x*_1, x*_2, \ldots, x*_n)$ of the matrix X, $1 \leq * \leq d$, the row sort mapping gives $\downarrow (x*_1, x*_2, \ldots, x*_n) = (x*_{p(1)}, x*_{p(2)}, \ldots, x*_{p(n)})$. where $p \in S_n$ is a permutation such that $x*_{p(1)} \leq x*_{p(2)} \leq \ldots, \leq x*_{p(n)}$ for $1 \leq * \leq d$.

For $d > 1$ a set of separating invariants $\beta_{A,B} : \mathbb{R}^{d\times n}/S_n \longrightarrow \mathbb{R}^{2nd}$ given by $\beta_{A,B}(X) = B(\beta_A(X))$ are suggested by Balan et al [2], which are the composition of sorting-based bilipschitz map,

$$\beta_A : \mathbb{R}^{d\times n}/S_n \longrightarrow \mathbb{R}^{D\times n} \textit{ and the linear map } B : \mathbb{R}^{D\times n} \to \mathbb{R}^{2nd}$$

Where $\beta_A(X) =\downarrow (AX)$ is a row-wise sorting combined with the linear map $A \in \mathbb{R}^{D\times d}$ and n, d, D are natural numbers. Balan et al provided some interesting properties of these maps which are summarised in the following theorem 3.0.1.

Theorem 3.0.1. *The permutation invariant maps $\beta_{A,B} : \mathbb{R}^{d\times n}/S_n \longrightarrow \mathbb{R}^{2nd}$ and $\beta_A : \mathbb{R}^{d\times n}/S_n \longrightarrow \mathbb{R}^{D\times n}$ defined above satisfies the following properties:*
(1) If for any matrix $A \in \mathbb{R}^{D\times d}$ the embedding β_A is separating then it is Bi-Lipschitz and the upper lipschitz constant is given by the largest singular value of the linear component A.
(2) If matrix $A \in \mathbb{R}^{D\times d}$ is a full spark matrix and $D = 1 + (d-1)n!$ then β_A is separating on $\mathbb{R}^{d\times n}$ and the lower Lipschitz constant is given by the smallest d-th singular value of all the $d \times d$ sub-matrices of the matrix A.
(4) If for $A \in \mathbb{R}^{D\times d}$ the map $\beta_A : \mathbb{R}^{d\times n} \to \mathbb{R}^{D\times n}$ is injective then for almost any linear map

$$B : \mathbb{R}^{D\times n} \to \mathbb{R}^{2nd}$$

the map $\beta_{A,B}(X) = B(\beta_A(X))$ is Bi-Lipschitz.

This construction is separating when the embedding dimension D is larger than the multiple of $n!$. But, a large embedding dimension leads to a computationally expensive model. Therefore, it is always desired to find a way to reduce the embedding dimension so that the space and time complexity of the model can be reduced. In further work by Dym and Gortler[7] it is proved that to achieve the separation for the above sorting-based embeddings, the required embedding dimension is $2nd + 1$ which is significantly lower than what is required in[2].

Following the discussion above, our work is motivated by the problems of achieving bilipschitz, separating permutation invariants $f : \mathbb{R}^{d\times n} \to \mathbb{R}^D$ for lower embedding dimensions than previously known in [2, 7] so that the mapping can be computed efficiently in practice.

Additionally, we would like the ratio M/m to be as close as possible to 1 to ensure the robustness of the model.

Our main results and findings are listed below.

3.1 Results:

1. Balan et al [2] said that algebraic embeddings defined by the sum of power function on $\mathbb{R}^{d\times n}$ are not Lipschitz. We provide a full proof of this fact for $d = 1$ and then generalize it by proving that any smooth permutation invariant function is not Bi-Lipschitz for $d \geq 1$. This is also true for functions of the form
$$\sum_{i=1}^{n} \psi_i(x)$$
even if they are only piecewise-smooth.

2. It is already known, for $d = 1$ sorting on $\mathbb{R}^{d\times n}$ is an isometry[2]. We added our own proof for this result.

3. We improved the bounds for the Lipschitz constants of the map $\beta_A(X) =\downarrow (AX)$. And showed that the Bi-Lipschitz constants $m(A)$ and $M(A)$ satisfy
$$M(A) = \sigma_1(A), \text{ and } \sigma_{d,k}(A) \leq m \leq \sigma_d(A)$$
Where $\sigma_1(A)$ is the largest singular value and $\sigma_d(A)$ is the d-th singular value of A. For every $k \geq n(d-1)$, $\sigma_{d,k}(A)$ is the smallest of all the k-th singular value of $k \times d$ sub-matrices $A(I) = (A_{ij})_{i\in I, 1\leq j\leq d}$ of A i.e.
$$\sigma_{d,k}(A) = \min_{I\subseteq\{1,\ldots,D\},|I|=k} \sigma_k(A(I)).$$

4. We proved that for the embedding dimension $D > n(d-1)$, the map β_A defined above which combines sorting with a linear mapping is separating and Bi-Lipschitz. Which improves upon the best-known lower bound of 2nd+1 from [7] and $1 + (d-1)n!$ from [2]. i.e.

Balan et al[2]	Dym and Gortler[7]	Our Result
$D = 1 + (d-1)n!$	$D = 2nd + 1$	$D > n(d-1) = 2nd+1-(nd+n+1)$

Table 1: Embedding dimension D

5. We performed some experiments to analyze the stability and accuracy of a model based on the sorting-based embedding scheme. Our experiments produced some interesting and exciting results. The existing model achieved high stability and accuracy for an embedding dimension which is much lower than what we require in our results.

4 Thesis Structure

The thesis is structured as follows:

Section 3: In this section, we have provided a brief introduction to our work, previous related work, and some of the mathematical background and definitions that are necessary to state our main goal and results.

Section 5: In this section, we discuss the algebraic embeddings (sum of power function) and devise a proof to show that these embeddings are not Bi-Lipschitz.

Section 6: In this section, we proved that in general, any permutation invariant function defined by the summation of smooth or piecewise smooth function is not Bi-Lipschitz for $d \geq 1$.

Section 7: In this section, we improved the embedding dimension required to prove the separation of the sorting-based permutation invariant embedding scheme. we also improved the bounds of the lipschitz constant of this embedding scheme for a lower embedding dimension than the previous required embedding dimensions.

Section 8: In this section, we provide the details of the experiment and experimental observations to analyze the stability and accuracy of the map β_A.

Section 9: Provides the conclusion of our work.

Section 10: This section discusses future research directions based on this work.

5 Algebraic Embedding (sum of the power function)

Polynomial (algebraic) embeddings are discussed by Balan et al in [2] by constructing their algebraic equivalent description which is the power sum function given below.

Power sum function:

$$f : \mathbb{R}^n \to \mathbb{R}^n,\ f(x) = \left(\sum_{l=1}^{n} x_l, \sum_{l=1}^{n} x_l^2 \cdots \sum_{l=1}^{n} x_l^n \right) \tag{1}$$

Lemma 5.0.1. The sum of the power function $f : \mathbb{R}^n \to \mathbb{R}^n$ mentioned above is permutation invariant and separating on $\mathbb{R}^n/S_n$.[2, 20]

We will show that the above algebraic embedding is not bilipschitz. To prove this result we will use the following Proposition. Proof of this proposition will be provided later on in section 7.1.

Proposition 5.0.2. *The sorting map* $\downarrow : \frac{\mathbb{R}^n}{S^n} \longrightarrow \mathbb{R}^n$, *is an isometry*[2]. *That is, for* $X, Y \in \frac{\mathbb{R}^n}{S^n}$, *and the metric* d *defined in definition (3.0.3), we have*

$$d(X, Y) = \| \downarrow (X) - \downarrow (Y) \|_2$$

Remark5: In other words, if $\Omega_n = \{(X_1, X_2, ...X_n) : X_1 \leq X_2 \leq X_3 ..... \leq X_n\} \subset \mathbb{R}^n$ Where all the elements in Ω_n are sorted in increasing order. then for d defined in definition (3.0.3),

$$d(X, Y) = ||X - Y||_2, \quad \forall X, Y \in \Omega_n$$

Lemma 5.0.3. The sum of power function $f : \mathbb{R}^n \to \mathbb{R}^n$ is not Bi-Lipschitz.

Proof. The power sum function $f : \mathbb{R}^n \to \mathbb{R}^n$ is defined as follows:

$$f(x) = \left(\sum_{i=1}^{n} x_i, \sum_{i=1}^{n} x_i^2, \ldots, \sum_{i=1}^{n} x_i^n \right)$$

From Lemma 5.0.1 we know that this map is permutation invariant and separating. To check whether the above function is Bi-Lipschitz, let's take a metric on $\mathbb{R}^n/\sim$ defined in definition 3.0.3, i.e.

$$d(X, Y) = \min_{\sigma \in S_n} \sqrt{\sum_{i=1}^{n} (x_i - y_{\sigma_{(i)}})^2} \tag{2}$$

The function f is Bi-Lipschitz with the constant $M \geq m > 0$ if,

$$m \leq \frac{||f(X) - f(Y)||_2}{d(X, Y)} \leq M \tag{3}$$

Where,

$$M = \sup_{X, Y \in \mathbb{R}^n \ d(X,Y) \neq 0} \frac{||f(X) - f(Y)||_2}{d(X, Y)} \tag{4}$$

$$m = \inf_{X, Y \in \mathbb{R}^n \ d(X,Y) \neq 0} \frac{||f(X) - f(Y)||_2}{d(X, Y)} \tag{5}$$

The Jacobian of the function f at a point $X \in \mathbb{R}^n$ is given by:

$$J_{f(X)} = \begin{pmatrix} 1 & 1 & 1 & \dots & 1 \\ 2x_1 & 2x_2 & 2x_3 & \dots & 2x_n \\ \\ \vdots & \vdots & \ddots & \vdots & \\ nx_1^{n-1} & nx_2^{n-1} & nx_3^{n-1} & \dots & nx_n^{n-1} \end{pmatrix}$$

In particular, Let $X = (x_1, x_2, ...x_n)$ s.t $x_1 = x_2$ and $x_1 \leq x_2 \leq ..... \leq x_n$ so that $X \in \Omega_n$. In this case, the first two columns of the above matrix are identical we have $det(J_{f(X)}) = 0$. Indeed, the vector

$$a = (1, -1, 0, 0, \dots, 0) \in \mathbb{R}^n$$

is in the null space of $J_{f(X)}$. We consider points of the form $X_\lambda = X - \lambda \cdot a$. Let's expand the function $f(X)$ using Taylor's expansion around the point X_λ:

$$\begin{aligned} f(X_\lambda) &= f(X) - \lambda J_{f(X)}.a + O(\|(\lambda.a)\|^2) \\ &= f(X) + O(\lambda^2). \end{aligned}$$

Recall that this last equality means that there exists some constant $C > 0$ such that for all small enough λ $\|f(X_\lambda) - f(X)\| \leq C\lambda^2$.

Since $X \in \Omega_n$, and for all small enough positive λ we have that X_λ is also in Ω_n. therefore by Proposition 5.0.2 along with the remark we have that

$$d(X_\lambda, X) = \|X_\lambda - X\|_2 = \lambda\|a\|_2 = \sqrt{2}\lambda.$$

Thus we can bound the expression in (3) by noting that for all $\lambda > 0$

$$m \leq \frac{||f(X_\lambda) - f(X)||_2}{|X_\lambda - X|} \leq \frac{C}{\sqrt{2}} \frac{\lambda^2}{\lambda} \longrightarrow 0 \quad as \quad \lambda \longrightarrow 0$$

Thus $m = 0$, and so the function $p : \mathbb{R}^n \longrightarrow \mathbb{R}^n$ is not Bi-Lipschitz. □

Therefore, on one-dimensional space power sum polynomials are not Bi-Lipschitz. In the following section, we will generalize the above result and show that *any* permutation invariant, separating function which is differentiable at a point $\mathbf{X} = (x_{i,j}) \in \mathbb{R}^{d \times n}$ having two identical columns $\mathbf{X_{i,1}} = \mathbf{X_{i,2}}$ along with one more additional condition will not be Bi-Lipshcitz. Let's define this problem more formally in the following section. A special case of this result was recently published in our preprint [1].

6 Continuously Differentiable Functions

Theorem 6.0.1. *Let $f : \mathbb{R}^{d\times n} \longrightarrow \mathbb{R}^D$ be a permutation invariant function which is differentiable at a point $\mathbf{X} = (x_{i,j})$ with column $(x_{i,1})_{1\le i\le d} = \mathbf{X_1} = \mathbf{X_2} = (\mathbf{x_{i,2}})_{\mathbf{1\le i\le d}}$ and each entry*

$$x_{i,2} \le x_{i,3} \le \ldots \le x_{i,n}, \quad x_{i,j} \in \mathbf{X_j}, \forall i = 1\ldots, d \text{ and } j = 1, \ldots n$$

Then f is not Bi-Lipschitz.

Proof. let's take $(x_{i,j}) = \mathbf{X} \in \mathbb{R}^{d\times n}$, Satisfying the condition $x_{i,1} = x_{i,2} \le x_{i,3} \le \ldots \le x_{i,n}, \forall i = 1\ldots, d$. Where $x_{i,1}, x_{i,2} \ldots, x_{i,n}$ denote the entries in the first, second, and n-th columns respectively.
For some infinitesimal small $t > 0$, let's define a function $g : \mathbb{R} \longrightarrow \mathbb{R}^{d\times n}$ s.t.

$$g(t) = \begin{pmatrix} x_{1,1} - t & x_{1,1} + t & x_{1,3} & \ldots & x_{1,n} \\ x_{2,1} - t & x_{2,1} + t & x_{2,3} & \ldots & x_{2,n} \\ \vdots & \vdots & \ddots & \vdots & \\ x_{d,1} - t & x_{d,1} + t & x_{d,3} & \ldots & x_{d,n} \end{pmatrix}$$

Denote the k-th column of $\mathbf{X}$ by $\mathbf{X_k} = (\mathbf{x_{i,k}})_{\mathbf{1\le i\le d}}$ and $\mathbf{1_d}$ the unit column of length d. then we can write
$g(t) = (\mathbf{X_1} - t\mathbf{1_d}, \mathbf{X_1} + t\mathbf{1_d}, \mathbf{X_3}, \ldots\ldots\mathbf{X_n})$ and $g(0) = (\mathbf{X_1}, \mathbf{X_1}, \mathbf{X_3}, \ldots\ldots\mathbf{X_n})$ therefore,

$$f(\mathbf{X_1} - t\mathbf{1_d}, \mathbf{X_1} + t\mathbf{1_d}, \mathbf{X_3}, \ldots\ldots\mathbf{X_n}) = f \circ g(t)$$

$$\implies f(\mathbf{X}) = f \circ g(0)$$

now using these $g(t)$ and $g(0)$ in the following expression we obtained,

$$\begin{aligned} m &= \inf_{\mathbf{X},\mathbf{Y}\in\mathbb{R}^{d\times n}, d(\mathbf{X},\mathbf{Y})\neq 0} \frac{||f(\mathbf{X}) - f(\mathbf{Y})||_2}{d(\mathbf{X},\mathbf{Y})} \\ &\le_{g(t),g(0)\in\mathbb{R}^{d\times n}, d(g(t),g(0))\neq 0} \frac{||f \circ (g(0+t)) - f \circ (g(0))||_2}{d(g(0+t), g(0))} \end{aligned} \tag{6}$$

As each $x_{i,j} \le x_{i,j+1}$ $\forall i = 1\ldots d$ *and* $\forall j = 1\ldots n$ such that $x_{i,j} \in \mathbf{X_j}$ *and* $x_{i,j+1} \in \mathbf{X_{j+1}}$, therefore using the remark5 for each i-th row of $g(t)$ and $g(0)$, we have

$$d(g(t)_i, g(0)_i) = \|g(t)_i - g(0)_i\|_F$$

where $g(t)_i$ and $g(0)_i$ denote the i-th rows of $g(t)$ and $g(0)$ respectively.

$$\begin{aligned} d(g(0+t), g(0)) &= \|(\mathbf{X_1} - t\mathbf{1_d}, \mathbf{X_1} + t\mathbf{1_d}, \mathbf{X_3}, \ldots\ldots\mathbf{X_n}) - (\mathbf{X_1}, \mathbf{X_1}, \mathbf{X_3}, \ldots\ldots\mathbf{X_n})\|_F \\ &= \|(-t\mathbf{1_d}, t\mathbf{1_d}, \mathbf{0_d}, \mathbf{0_d}, \ldots\ldots\mathbf{0_d})\|_F \quad \textit{where } \mathbf{0_d} \textit{ is a zero column vector} \\ &= \sqrt{2d}t \ldots\ldots (\because X_i \in \mathbb{R}^d) \end{aligned} \tag{7}$$

Now to simplify the numerator let's define a function:

$$\psi : \mathbb{R} \longrightarrow \mathbb{R}^D$$

such that

$$\psi(t) = f \circ g(t)$$

Since f is S_n invariant, therefore, $\psi(-t) = f \circ (g(-t)) = f \circ (g(t)) = \psi(t)$ which shows that ψ is a symmetric function of $\mathbb{R}$.

By the differentiability of f at $\mathbf{X} = (\mathbf{X_1}, \mathbf{X_1}, \mathbf{X_3} \ldots, \mathbf{X_n})$, ψ is differential at $t = 0$, we have

$$\left| \frac{\psi(t) - \psi(0) - \psi'(0)t}{t} \right| \xrightarrow{t \to 0} 0 \ \textit{and} \ \left| \frac{\psi(-t) - \psi(0) - \psi'(0)(-t)}{-t} \right| \xrightarrow{t \to 0} 0 \quad (\because \psi(t) = \psi(-t)) \tag{8}$$

We can write

$$\begin{aligned} \frac{|\psi(t) - \psi(0)|}{|t|} &= \frac{|\psi(t) - 2\psi(0) + \psi(-t)|}{2\,|t|} \\ &= \frac{|\psi(t) - \psi(0) - \psi'(0)t + \psi(-t) - \psi(0) + \psi'(0)t|}{2\,|t|} \\ &\xrightarrow{t \to 0} 0 \quad \textit{using (8)} \end{aligned} \tag{9}$$

equation (7) and (9) implies,

$$\frac{|\psi(t) - \psi(0)|}{\sqrt{2} \cdot d\,|t|} \xrightarrow{t \to 0} 0 \tag{10}$$

Therefore,

$$m \leq_{g(t), g(0) \in \mathbb{R}^{d \times n} \ ||g(t) - g(0)||_F \neq 0} \frac{||f \circ (g(0+t)) - f \circ (g(0))||_2}{||g(0+t) - g(0)||_F} \xrightarrow{t \to 0} 0 \tag{11}$$

Thus, $m \to 0$ as $t \to 0$.

We obtained $m = 0$ but by definition(3.0.5) the Bi-Lipschitz constants m *and* M must be non-zero. This shows every permutation invariant function that satisfies the conditions given in the theorem is not Bi-Lipschitz on $\mathbb{R}^{d \times n}$. □

Next, we analyze the behavior of the piece-wise smooth functions of the form $f(X) = \sum_{i=1}^{n} \psi_i(X)$ which are permutation invariant. Such functions are common in graph and set neural networks, where ψ is typically a fully connected neural network. [1, 19, 20, 17, 18]

Using the above theorem we will show that these functions are not Bi-Lipschitz. Let's state this in the following Corollary.

Corollary 6.0.2. *For every $\psi : \mathbb{R}^d \to \mathbb{R}^D$ which is differentiable in at least one point, the function $f : \mathbb{R}^{d \times n} \longrightarrow \mathbb{R}^D$ s.t $f(\mathbf{X}_1, \ldots, \mathbf{X}_n) = \sum_{j=1}^{n} \psi(\mathbf{X}_j)$ is not Bi-Lipschitz. Where $\mathbf{X}_k = (x_{i,k})_{1 \leq i \leq d}$ denote the k-th column of $\mathbf{X} \in \mathbb{R}^d$ and $1 \leq k \leq n$.*

Proof. Take $\mathbf{X}_1 \in \mathbb{R}^d$ to be a point on which ψ is differentiable. Then f is differentiable at $X = (\mathbf{X}_1, \mathbf{X}_1, \ldots, \mathbf{X}_1)$ and therefore f is not Bi-Lipschitz by Theorem 6.0.1 □

In the next section, we'll analyze the Bi-Lipschitz properties of sorting-based embedding schemes.

7 Sorting-Based Embedding Schemes

Below is the description of the sorting map defined on $\mathbb{R}^n$.

Definition 7.0.1 (Sorting)**.** Sorting, denoted by $\downarrow$, arranges the elements of input in increasing order. For the permutation group S^n the map

$$\downarrow : \frac{\mathbb{R}^n}{S^n} \longrightarrow \mathbb{R}^n,$$

is defined by sending an equivalence class $X = [X]$ to the unique representative $X' = \downarrow([X])$ in the equivalence class which satisfies $X'_1 \leq X'_2 \leq X'_3, ... \leq X'_n$.
where X can be any choice of a representative of $[X] \in \frac{\mathbb{R}^n}{S^n}$.

Remark: By definition sorting is permutation invariant and separates the points of $\mathbb{R}^n/S_n$.

7.1 One Dimensional Sorting

Sorting defined on $\mathbb{R}^{d\times n}$ is an isometry for dimension $d = 1$ and this is already known in [6] but it was difficult to trace down the proof so we supply our own proof for this elementary fact.

Proposition 7.1.1. *The Sorting map* $\downarrow : \frac{\mathbb{R}^n}{S^n} \longrightarrow \mathbb{R}^n$ *is an isometry. That is, for* $X, Y \in \frac{\mathbb{R}^n}{S^n}$*, and a metric* d *defined in definition (3.0.3), following holds,*

$$d(X, Y) = \| \downarrow (X) - \downarrow (Y)\|$$

Proof. let's take a set $\Omega_n = \{(X_1, X_2, ...X_n) : X_1 \leq X_2 \leq X_3 ..... \leq X_n\} \subset \mathbb{R}^n$. Where all the elements in Ω are sorted in increasing order. To prove the lemma we will show that for d defined in definition 3.0.3,

$$d(X, Y) = ||X - Y||, \quad \forall X, Y \in \Omega_n \quad .....(2)$$

Let's start proving for $n = 2$ and then for an arbitrary value of n.
Case 1: for $n = 2$
$\Omega_2 = \{(X_1, X_2) : X_1 \leq X_2\}$ and suppose that $X,\ Y \in \Omega_2$ s.t $X = (X_1, X_2)$ *and* $Y = (Y_1, Y_2)$.

If the identity permutation attains the minimum in (3.0.3) then $d(X, Y) = \|X - Y\|$ as required. Assume by contradiction that the non-identity permutation $\sigma \in S_2$ which satisfies $Y_{\sigma_{(1)}} = Y_2$ *and* $Y_{\sigma_{(2)}} = Y_1$. attains the minimum in the definition of $d(X, Y)$. Then

$$\begin{aligned}
||X - Y||^2 - d(X, Y)^2 &= (X_1 - Y_1)^2 + (X_2 - Y_2)^2 - (X_1 - Y_{\sigma_{(1)}})^2 - (X_2 - Y_{\sigma_{(2)}})^2 \\
&= (X_1 - Y_1)^2 + (X_2 - Y_2)^2 - (X_1 - Y_2)^2 - (X_2 - Y_1)^2 \\
&= 2X_1(Y_2 - Y_1) - 2X_2(Y_2 - Y_1) \\
&= 2(X_1 - X_2)(Y_2 - Y_1) \leq 0
\end{aligned}$$

which is a contradiction to the fact that $d(X, Y)$ is achieved by σ.

Case 2: n is arbitrary,
$\Omega_n = \{(X_1, X_2, X_3....., X_n) : X_1 \leq X_2 \leq X_3 ... \leq X_n\} \subseteq \mathbb{R}^n$
let $X, Y \in \Omega_n$, and assume by contradiction that there is a non-identity permutation $\sigma \in S_n$ such that

$$||X - Y|| > d(X, Y) = \|X - \sigma(Y)\|$$

As σ is a non-identity permutation, there is at least one pair satisfying $Y_{\sigma_{(j)}} \geq Y_{\sigma_{(k)}}$ for $j < k$. If there is only one pair that is not fixed by σ then this case reduces to case1 and we are done. But

if there is more than one pair satisfying $Y_{\sigma_{(j)}} \geq Y_{\sigma_{(k)}}$ for $j < k$ then take a permutation $\tau_1 \in S_n$ which swaps two element such that $\tau_1(j) = k$ *and* $\tau_1(k) = j$ leaving other elements unchanged.

Composing with σ construct a permutation $\tau_1 \circ \sigma$ which is a better permutation than σ satisfying

$$\|X - \tau_1 \circ \sigma(Y)\| \leq \|X - \sigma(Y)\| < \|X - Y\|$$

We will apply the same process as above to swap the other elements to get a better permutation that gives a norm better than the previous one. suppose that there are $r-1$ remaining pairs such that $Y_{\sigma_{(j)}} \leq Y_{\sigma_{(k)}}$ for $j < k$ then consider the permutations $\tau_2, \tau_3, \ldots, \tau_r$ such that each σ_i swap one pair and leave the rest unchanged. Composing all these permutations we get $\tau_r \circ \tau_{r-1} \circ \ldots \tau_2 \circ \tau_1 \circ \sigma$ such that

$$\tau_r \circ \tau_{r-1} \circ \ldots \tau_2 \circ \tau_1 \circ \sigma(Y) = Y$$

Which is identity permutation satisfying

$$\|X - \tau_r \circ \tau_{r-1} \circ \ldots \tau_2 \circ \tau_1 \circ \sigma(Y)\| \leq \|X - \tau_1 \circ \sigma(Y)\| \leq \|X - \sigma(Y)\|$$

Therefore, the identity permutation achieves $d(X, Y)$. From the above discussion, we conclude that the sorting operation is an isometry. $\square$

From the above remark and proposition, it is clear that the map $\downarrow: \frac{\mathbb{R}^n}{S^n} \longrightarrow \mathbb{R}^n$, is separating, permutation invariant, and isometry (Bi-Lipschitz). Next, we will analyze the sorting-based permutation invariant mapping on $\mathbb{R}^{d \times n}$ for $d > 1$.

Note that, here we will use the usual row-wise sorting where all the rows of the matrix are sorted independently in increasing order.

7.2 Multidimensional Sorting

We consider mappings $\beta_A : \mathbb{R}^{d\times n}/S_n \longrightarrow \mathbb{R}^{D\times n}$ of the form $\beta_A(X) =\downarrow (AX)$. Here $\downarrow$ represents the operation of sorting each row of the matrix independently as defined in definition 3.0.7.

Clearly, this map is permutation invariant. In the following theorem, we will show that the above map is separating whenever $A \in \mathbb{R}^{D\times d}$ is the full spark and $D > n(d-1)$. This improves on the cardinality of D required for separation in [7, 2]. It is proved by Balan [2] injectivity of β_A implies Bi-Lipschitz. Therefore if we prove that β_A is separating for $D > n(d-1)$ then it is Bi-Lipschitz.

Let's introduce the full spark matrix in the following definition.

Definition 7.2.1 (Full Spark Matrix)**.** Let $A \in \mathbb{R}^{D\times d}$, where $D > d$. If every $d \times d$ sub-matrix of A is invertible then the matrix A is said to be a full spark matrix.

Theorem 7.2.1. *Let n, $d \geq 2$ and $D > n(d-1)$ be natural numbers, and let $A \in \mathbb{R}^{D\times d}$ be a full spark matrix, then the mapping*

$$\beta_A : \mathbb{R}^{d\times n} \longrightarrow \mathbb{R}^{D\times n}$$

is permutation invariant and separating.

Proof. Fix D, d s.t $D \geq d \geq 2$ and let $A \in \mathbb{R}^{D\times d}$ be a full spark matrix (recall this means that all the $d \times d$ submatrices of A are of rank d).

Let $n = n(D, d)$ be the smallest natural number, such that there exist $X, Y \in \mathbb{R}^{d\times n}$ which are not equal up to permutation, but $\beta_A(X) =\downarrow (AX) =\downarrow (AY) = \beta_A(X)$. We need to show that $D \leq n(d-1)$.

By assumption, each column X_i is not equal to any column Y_j. otherwise, we would obtain a contradiction to the minimality of n by removing these columns from X and Y and obtaining smaller point clouds with the same properties. This implies that each pair AX_i and AY_j have at most d-1 entries in common (since A is full spark).

Fix i. Each of the D entries of AX_i is equal to the entries of one of $AY_j \quad j = 1, 2, 3.....n$ as $\downarrow (AX) =\downarrow (AY)$. Since we saw that each column AY_j has at most $d-1$ entries in common with AX_i, it follows that $D \leq n(d-1)$. □

Theorem 3.10 in Balan's paper [2] says that injectivity implies Bi-Lipschitz but doesn't specify the Lipschitz constants. To determine the stability of β_A we study its Bi-Lipschitz constants. The constants we prove at least the lower Lipschitz constant is also specified therein [2] but only when D is huge which we proved for a relatively very small value of D than in [2].

To prove the bounds on Lipschitz constants, we introduce the following notation: For $I \subseteq \{1, \ldots, D\}$ we denote $A(I) = (A_{ij})_{i\in I, 1\leq j\leq d}$. We denote the d-th singular value of $A(I)$ by $\sigma_d(A(I))$ and for $k \geq d$ we define $\sigma_{d,k}$ to be the minimal d-th singular value, of all possible $k \times d$ submatrices of A, that is:

$$\sigma_{d,k}(A) = \min_{I\subseteq\{1,\ldots,D\}, |I|=k} \sigma_d(A(I)).$$

We denote the maximal singular value of A by $\sigma_1(A)$. To prove the relationship between the stability constants of β_A and $\sigma_{d,k}$ we will need the following definitions:

Definition 7.2.2 (Doubly Stochastic Matrix)**.** A square matrix $A = (a_{ij})$ is doubly stochastic if all its entries are non-negative real numbers, and all rows and columns sum to one.

We will use the following two known theorems to prove our further results:

Theorem 7.2.2 (Birkhoff-Von-Neumann [4])**.** *If $A \in \mathbb{R}^{n\times n}$ is a doubly stochastic matrix, then there exist some N and non-negative numbers $t_1, t_2, \ldots, t_N \geq 0$, s.t $\sum_{i=1}^{N} t_i = 1$, and permutation matrices $P_1, P_2, \ldots, P_N$, such that $A = t_1P_1 + t_2P_2 + \cdots + t_NP_N$ (A is a convex combination of permutation matrices).*

Theorem 7.2.3 (Caratheodory Theorem [5])**.** *Let $S \subset \mathbb{R}^d$, if $x \in Conv(S) \subset \mathbb{R}^d$ then x can be written as a convex combination of $d+1$ extremal points of S.*

Remark: Let $DS_n = \{A \in \mathbb{R}^{n\times n} : \sum_{i=1}^{n} a_{ij} = 1, \sum_{j=1}^{n} a_{ij} = 1, a_{ij} \geq 0\}$ be the set of all the doubly stochastic matrices. It is a compact convex polyhedron in Euclidean $n^2 - space$ with dimension $(n-1)^2$ and the $n \times n$ permutation matrices as the extreme points(vertices). (see [11]). Then using the theorem 7.2.2 and 7.2.3 every doubly stochastic matrix can be written as the convex combination of at most $(n-1)^2 + 1$ permutation matrices.

Theorem 7.2.4. *Let $A \in \mathbb{R}^{D\times d}$ be a full spark matrix, and assume that $k \geq d \geq 2$ and $D \geq k\left[(n-1)^2 + 1\right]$, then*

$$\|\beta_A(X) - \beta_A(Y)\|_F \geq \sigma_{d,k}(A)d(X,Y)$$

Proof. Let's consider two matrices $X, Y \in \mathbb{R}^{d\times n}$, and let us denote $\bar{X} = AX, \bar{Y} = AY$. Let the i-th row and j-th column of a matrix $X \in \mathbb{R}^{d\times n}$ be denoted by $\bar{X}_{i*}$ and $\bar{X}_{*j}$ respectively. There exist permutations $\sigma_1, \ldots, \sigma_D \in S_n$ such that

$$\begin{aligned}\|\beta_A(X) - \beta_A(Y)\|_F^2 &= \sum_{i=1}^{D} \| \downarrow (\bar{X}_{i*}) - \downarrow (\bar{Y}_{i*})\|_2^2 = \sum_{i=1}^{D}\sum_{j=1}^{n} |\bar{X}_{ij} - \bar{Y}_{i\sigma_i(j)}|^2 \\ &= \sum_{j=1}^{n}\sum_{\ell=1}^{n}\sum_{i\in I_{j,\ell}} |\bar{X}_{ij} - \bar{Y}_{i\ell}|^2 \end{aligned} \tag{12}$$

where we use the notation

$$I_{j,\ell} = \{i|\, \sigma_i(j) = \ell\}.$$

Note that for fixed j we have $\sum_\ell |I_{j,\ell}| = D$, and for fixed ℓ we have $\sum_j |I_{j,\ell}| = D$. We deduce

Lemma 7.2.5. There exists a permutation $\sigma \in S_n$ such that

$$|I_{j\sigma(j)}| \geq k, \forall j = 1, \ldots, n$$

Proof. Let $S \in \mathbb{R}^{n\times n}$ be defined by $S_{j\ell} = |I_{j\ell}|$. Then $\frac{1}{D}S$ is a doubly stochastic matrix, that is, its entries are all non-negative and its rows and columns sum to one. By the Birkhoff-Von-Neumann Theorem coupled with the Caratheodory Theorem, we can write $\frac{1}{D}S$ as a convex combination of $N = (n-1)^2 + 1$ permutation matrices

$$\frac{1}{D}S = \sum_{s=1}^{N} \theta_s P^{(s)},$$

where θ_i are non-negative and sum to one. Let t be such that $\theta_t \geq \theta_s, \forall s$. It follows that $\theta_t \geq \frac{1}{N}$. Let σ be the permutation corresponding to $P^{(t)}$, such that

$$P_{j\ell}^{(t)} = \begin{cases} 1 & \text{if } \ell = \sigma(j) \\ 0 & \text{if } otherwise \end{cases}$$

Then we have for all j

$$0 \leq \frac{1}{D} S_{j\sigma(j)} - \theta_t P^{(t)}_{j\sigma(j)} = \frac{1}{D} S_{j\sigma(j)} - \theta_t \leq \frac{1}{D} S_{j\sigma(j)} - \frac{1}{N}$$

and so for all j,

$$|I_{j\sigma(j)}| = S_{j\sigma(j)} \geq \frac{D}{N} \geq k$$

□

Returning to (12) we obtain

$$\begin{aligned}
\|\beta_A(X) - \beta_A(Y)\|_F &= \sum_{j=1}^{n} \sum_{\ell=1}^{n} \sum_{i \in I_{j,\ell}} |\hat{X}_{ij} - \hat{Y}_{i\ell}|^2 \\
&\geq \sum_{j=1}^{n} \sum_{i \in I_{j,\sigma(j)}} |\bar{X}_{ij} - \bar{Y}_{i\sigma(j)}|^2 \\
&= \sum_{j=1}^{n} \|A(I_{j,\sigma(j)})X_j - A(I_{j,\sigma(j)})Y_{\sigma(j)}\|_2^2 \\
&\geq \sigma_{d,k}(A)^2 \sum_{j=1}^{n} \|X_j - Y_{\sigma(j)}\|_2^2 \\
&\geq \sigma_{d,k}(A)^2 d^2(X, Y)
\end{aligned}$$

□

The following theorem shows that the optimal upper Lipschitz bound M for β_A is always simply $\sigma_1(A)$, the largest singular value of A (which is also its operator norm). In the other direction, the smallest singular value σ_d of A is an upper bound for the optimal lower Lipschitz bound m. This is true unconditionally for all A. When the condition of the previous theorem holds we also have a lower bound on m,

$$\sigma_{d,k}(A) \leq m \leq \sigma_d(A)$$

Theorem 7.2.6. *Let $A \in \mathbb{R}^{D\times d}$ and $X \in \mathbb{R}^{d\times n}/S_n$. then the mapping $\beta_A : \mathbb{R}^{d\times n}/S_n \longrightarrow \mathbb{R}^{D\times n}$ given by*

$$\beta_A(X) =\downarrow (AX)$$

satisfies,

$$m\|X - Y\|_F \leq \|\beta_A(X) - \beta_A(Y)\|_F \leq M\|X - Y\|_F.$$

Where $M = \sigma_1(A)$ and $m \leq \sigma_d(A)$. Moreover, there exists X, Y for which there is equality on the right side.

Proof. First, we will show that,

$$\|\beta_A(X) - \beta_A(Y)\|_F \leq M\|X - Y\|_F \tag{13}$$

This proof already appeared in [2] and we repeat it here for completeness. Let $X, Y \in \mathbb{R}^{d\times n}$ s.t $d(X, Y) \neq 0$. Let $a_1 \dots a_D$ denote the rows of the matrix A. That is

$$A = \begin{bmatrix} a_1 \\ \hline a_2 \\ \hline \vdots \\ \hline a_D \end{bmatrix}.$$

Then there exist the permutation matrices $P^*, P_1, P_2 \dots P_D$ *and* $\Pi_1 \dots \Pi_D \in S_n$ such that, For any $X, Y \in \mathbb{R}^{d\times n}/S_n$ we have

$$d(X, Y) = \|X - YP^*\|_F$$

and

$$\beta_A(X) =\downarrow A(X) = \begin{bmatrix} a_1XP_1 \\ \hline a_2XP_2 \\ \hline \vdots \\ \hline a_DXP_D \end{bmatrix} \text{ and } \beta_A(Y) =\downarrow A(Y) = \begin{bmatrix} a_1Y\Pi_1 \\ \hline a_2Y\Pi_2 \\ \hline \vdots \\ \hline a_DY\Pi_D \end{bmatrix}$$

Therefore,

$$\|\beta_A(X) - \beta_A(Y)\|_F^2 = \sum_{i=1}^{D} \|a_iXP_i - a_iY\Pi_i\|_2^2 = \sum_{i=1}^{D} \|a_i(XP_i\Pi_i^T - Y)\|_2^2$$

Since sorting is the optimal permutation therefore, we have

$$\|a_iXP_i - a_iY\Pi_i\|_2 = min_{P\in S_n}\|a_iX - a_iYP\|_2$$

Therefore, we have

$$\|a_iXP_i - a_iY\Pi_i\|_2 \leq \|a_iX - a_iYP^*\|_2$$

Thus

$$\|\beta_A(X) - \beta_A(Y)\|_F^2 \leq \sum_{i=1}^{D} \|a_i X - a_i Y P^*\|_2^2 = \|A(X - YP^*)\|_F^2 \leq \sigma_1(A)^2 \|X - YP^*\|_F^2$$

where $M = \|A\|_{op} = \sigma_1(A)$.Therefore,

$$\|\beta_A(X) - \beta_A(Y)\|_F \leq M d(X, Y)$$

This bound is attained when the considered matrices are with repeated columns: Let X^* and Y^* be such that $X^* = (X, X....., X)$ and $Y^* = (Y, Y....., Y)$. where X, Y are vectors in $\mathbb{R}^d$, and $X - Y$ is a vector of unit norm for which $\|A(X - Y)\|_2$ is maximal, that is $\|A(X - Y)\|_2 = \|A\|_{op}$. Since the columns of X^* are all identical therefore,

$$\beta_A(X^*) =\downarrow (AX^*) = AX^*$$

Thus,

$$\begin{aligned}
\|\beta_A(X^*) - \beta_A(Y^*)\|_F^2 &= \|A(X^*) - A(Y^*)\|_F^2 \\
&= \|A(X^* - Y^*)\|_F^2 \\
&= \|A\|_{op}^2 \sum_{i=1}^{n} \|(X - Y)\|_2^2 \\
&= n\|A\|_{op}^2
\end{aligned}$$

The distance between the above-selected matrices is given by,

$$\begin{aligned}
d(X^*, Y^*)^2 &= min_{P \in S_n} \|X^* - PY^*\|_F^2 \\
&= \|X^* - Y^*\|_F^2 \\
&= n\|X - Y\|_2^2 = n
\end{aligned}$$

Using this observation we obtained,

$$\begin{aligned}
\|\beta_A(X^*) - \beta_A(Y^*)\|_F &= \|A\|_{op} d(X^*, Y^*) \\
&= M \times d(X^*, Y^*)
\end{aligned}$$

where, $M = \|A\|_{op} = \sigma_1(A)$.

Therefore the inequality in the expression $\|\beta_A(X) - \beta_A(Y)\|_F \leq M d(X, Y)$ in (13) is in fact equality for $X = X^*$ *and* $Y = Y^*$.

We now prove that $m \leq \sigma_d(A)$. For some constant vectors $X, Y \in \mathbb{R}^d$ such that $\|X - Y\|_2 = 1$ and $\|A(X - Y)\|_2$ attains the minimal possible value $\sigma_d(A)$, we take X^*, Y^* to be the constant matrices with the columns X *and* Y as considered before. We then have

$$\begin{aligned}
m^2 &= \inf_{X \not\sim Y} \frac{\|\beta_A(X) - \beta_A(Y)\|_F^2}{d(X, Y)^2} \\
&\leq \frac{\|\beta_A(X^*) - \beta_A(Y^*)\|_F^2}{d(X^*, Y^*)^2} \\
&= \frac{\|A(X^*) - A(Y^*)\|_F^2}{\|X^* - Y^*\|_F^2} \\
&= \frac{n\|A(X - Y)\|_2^2}{n\|X - Y\|_2^2} \\
&= \sigma_d^2
\end{aligned}$$

Thus, $m \leq \sigma_d$.

$\square$

8 Experimental Observations

We performed two experiments to examine the quality of the sorting-based embedding β_A. (i) The first experiment empirically estimates the Lipschitz stability constants of β_A. and (ii) The second measures the accuracy of the β_A based model for a synthetic permutation-invariant binary classification task.

(i) Stability of the model: To check the stability of β_A empirically, we randomly generated a matrix $A \in \mathbb{R}^{D\times d}$, which embeds the input data into a D dimensional space, and then randomly generated $K = 1000$ pairs of point clouds (X_k, X'_k). We then estimated m, M through

$$\tilde{M} = \max_{1\leq k\leq K} \frac{\|\beta_A(X_k) - \beta_A(X'_k)\|_F}{d(X_k, X'_k)}, \text{ and } \tilde{m} = \min_{1\leq k\leq K} \frac{\|\beta_A(X_k) - \beta_A(X'_k)\|_F}{d(X_k, X'_k)}$$

The outcome of this experiment depends on the hyperparameters d, n (the point cloud dimensions) and D (the embedding dimension of β_A). In Figure 1 we examine the dependency of the ratio $\tilde{M}/\tilde{m}$ on each one of these parameters when the other two are held fixed. In all cases, we run the experiment independently 1000 times and report the average $\tilde{M}/\tilde{m}$ value.

This experiment gives the following observations.

- Variation in the stability w.r.t to the number of points n in the point cloud: In this experiment, we fixed the embedding dimension D and dimension of the input dataset d at 3 and performed the experiment for each $n \in \{5, 10, 15, 20, 25, 30, 35, 40, 45, 50\}$. In this setting the values of $\tilde{M}/\tilde{m}$ w.r.t n are shown in graph(i) of Figure 1. For small value $n \leq 10$ the ratio $\tilde{M}/\tilde{m}$ is more than 2.5, but as the number n increases to 20 this ratio drops abruptly to less than 2.25. For values more than $n = 20$, it remains approximately at 2. We note that this result is somewhat surprising, as the given embedding dimension $D > n(d-1)$ which is theoretically required for injectivity grows with n. This experiment suggests that the empirical stability ratio can be rather low even when D is significantly smaller than $n(d-1)$.

  In the next experiment, we fixed the size of the input point cloud and plot the variations w.r.t other parameters:

- Variation in the stability w.r.t to input dimension d: In this experiment, we fixed the number of points in the input point cloud at $n = 100$ and the embedding dimension at $D = 3$. The ratio $\tilde{M}/\tilde{m}$ for each $d \in \{2, 3, 4, 5 \ldots 15\}$ is calculated and the results of this experiment are plotted in graph(iii) of Figure 1. From this graph, it is observed that With the increase in the values of d the ratio M/m shows a sudden increase and decrease with an irregular zig-zag pattern so the dependence of stability on d is not clear from here.

- Variation in the stability w.r.t to the parameter D: We fixed the number of points in the point cloud at $n = 100$ and input dimension at $d = 3$. With this setting the value of ratio $\tilde{M}/\tilde{m}$ for each of $D \in \{5, 10, 20, 30, 40, 50, 60, 70, 80, 100\}$ is plotted in the *graph*(v) of Figure 1. For the small value of D the ratio, M/m is more than 2.8, and, it decreases abruptly for $5 < D < 20$. After that, there is a decrease in the ratio $\tilde{M}/\tilde{m}$ as the value of D increases and attains a value less than 2 for all the values of $D > 50$. This experiment shows the improvement in the stability of the model with the increase in the embedding dimension.

  Next, we will run a similar experiment to analyze the accuracy of permutation invariant models based on β_A in solving a synthetic permutation invariant binary classification task.

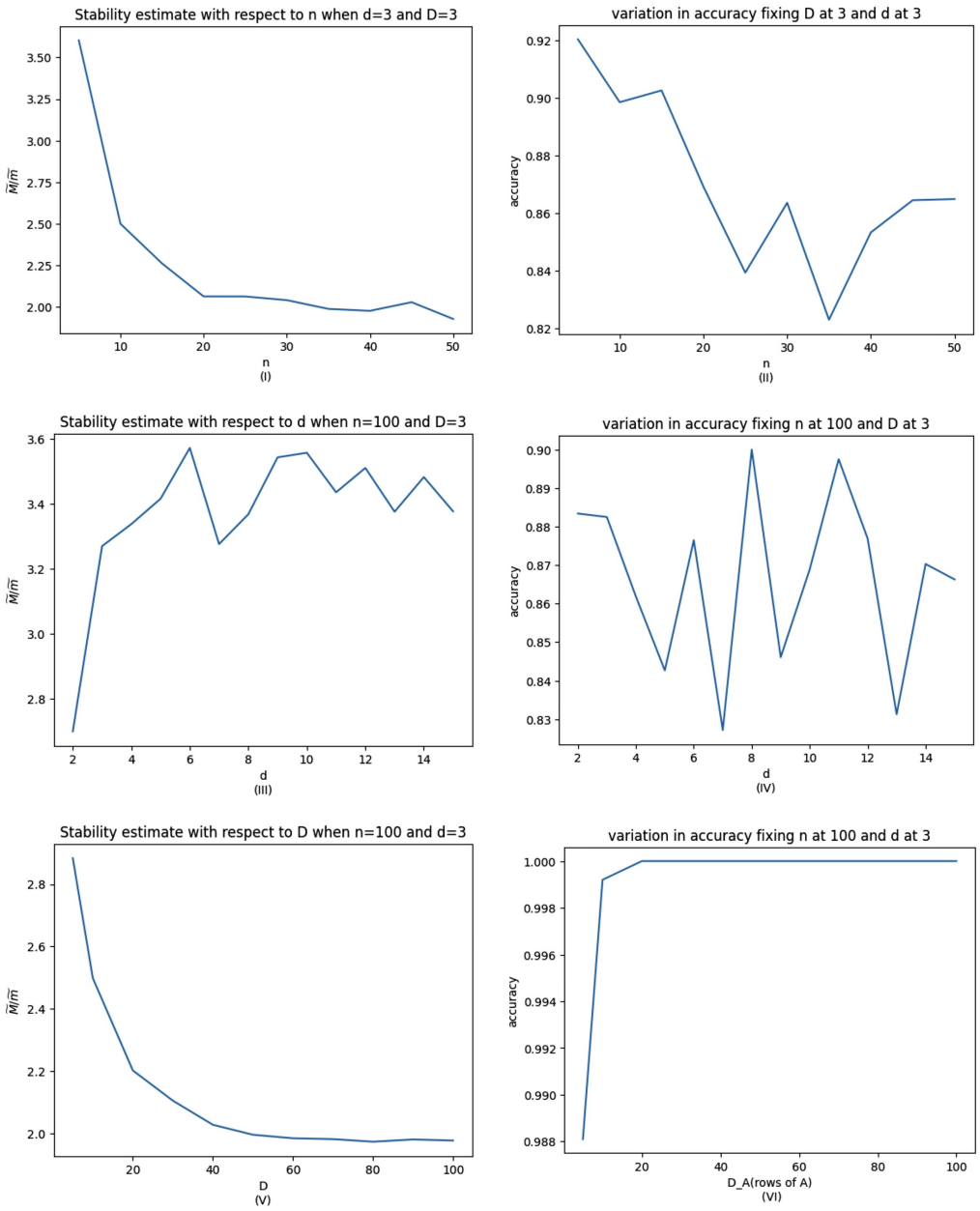

Figure 1: stability and accuracy of sorting-based embedding Scheme

Accuracy of a sorting-based permutation invariant model: We consider the following simple synthetic permutation invariant binary classification task. We randomly generated two lines

$$L_1 = \{(1-t)X^{(1)} + tY^{(1)} | \, t \in [0,1]\}, L_2 = \{(1-t)X^{(2)} + tY^{(2)} | \, t \in [0,1]\},$$

by randomly choosing the endpoints $X^{(1)}, Y^{(1)}, X^{(2)}, Y^{(2)} \in \mathbb{R}^{d \times n}$. We consider a synthetic learning problem where we are given a point cloud X from either L_1 or L_2, corrupted by some permutation matrix P, and the goal of the learning process is to discover whether XP came from L_1 or L_2. We note that this problem is permutation invariant by construction.

We address this problem by first applying β_A to the input point cloud X and then applying a fully connected network with three fully connected layers of width $1024, 512, 256$ respectively and finally logsoftmax in the output layer. The learning rate of the algorithm is fixed at 0.001,

The outcome of this experiment, and the dependence of accuracy on the parameters n, d, D, are shown in the right-hand side of Figure 1. The values in the graph are averaged over ten experiments performed by varying the different parameters.

- Variation of accuracy w.r.t. n: In the previous experiment, We saw that with the increase in the number of points n in the input point cloud, the ratio $\tilde{M}/\tilde{m}$ decreases i.e. the stability of the model increases. Surprisingly, in this experiment the classification accuracy *decreases* with the increase in the size of the input dataset.

- Variation of accuracy with respect to d: In the next experiment, we considered the effect of changing the input dimension d. In this case, the accuracy with respect to d shows an irregular increase and decrease, which is consistent with our outcome in the stability experiment.

- Variation of accuracy with respect to D: In this experiment, we fixed n and d, and we considered variations in the embedding dimension D. We see that there is a continuous improvement in the accuracy of the model. For the embedding dimension $D < 20$ the model achieves an accuracy of more than 99 percent. Further for all the values of $D > 20$, the model achieves an accuracy of 100 percent, which is depicted in $graph(vi)$ of Figure 1. This is consistent with our results in the stability experiment, where increasing D led to improved stability.

Therefore, empirically we found that the embedding dimension plays an important role in the stability and accuracy of the model. We have found that increasing the embedding dimension D consistently improves both the accuracy and the stability of the model. The dependence of stability and accuracy on n and d are less clear and are worthy of further investigation.

9 Conclusion

Let's summarize all results and findings: In this work, we analyzed the injectivity and stability of permutation invariant embedding schemes based on algebraic and sorting-based functions.

Our main results are:

- Every smooth separating function is not Bi-Lipscchitz when $d \geq 1$. This means that they do not preserve the distances between points in the embedding space. This can lead to problems such as the model assigning similar embeddings to two different inputs, which can cause errors in downstream tasks.

- we also proved that the piecewise smooth functions of the form

$$\sum_{i=1}^{n} \psi_i(x)$$

  are not Bi-lipschitz for $d \geq 1$.

- Sorting combined with linear map *i.e* $X \mapsto \beta_A(X) =\downarrow (AX)$ is always Bi-Lipschitz. We showed that the Lipschitz constants $m(A)$ and $M(A)$ of the map β_A satisfy

$$M(A) = \sigma_1(A), \text{ and } \sigma_{d,k}(A) \leq m \leq \sigma_d(A)$$

  Where $\sigma_1(A)$ is the largest singular value of A and $\sigma_{d,k}(A)$ is the smallest of all the d-th singular value of $d \times k$ sub-matrices of A. This was shown for $D \sim n^2 d$ measurements, improving upon $D > n!$ assumptions needed for similar results in [2].

- We managed to reduce the embedding dimension needed for injectivity to $D > n(d-1)$, improving upon the $D > 2nd$ bound from [7] and the significantly higher bound $D > n!$ in [2].

- Experiments with the model based on sorting-based embedding β_A on a synthetic dataset confirmed our findings and also produced some new insights into the stability and accuracy of the model. On our synthetic dataset, the model achieved high stability and accuracy even when the embedding dimension was very low compared to our theoretical result for the general data manifold. From these empirical results of the experiment, we found that the stability and accuracy of the model increases with the increase in the embedding dimension D keeping other parameters fixed, e.g. $n = 100$ and $d = 3$.

- We found that the stability of these embeddings increases with the size of the input dataset (synthetic).

- Surprisingly, we found that there is no clear relationship between the input size or dimension and the accuracy of the model. This suggests that the accuracy of the embeddings is not as dependent on the size or dimension of the input dataset as it is on the value of the embedding dimension.

  We hope these results and observations will assist the researchers in getting more insights about the sorting-based embedding schemes and encourage follow-up research in this field.

10 Further Direction

- It is always desired to optimize the embedding dimension of the network to reduce the overall computation cost of the model. We proved that networks with sorting-based embedding β_A are separating for $D > n(d-1)$ where n is the number of data points and d is the input dimension. However, in our synthetic experiments, we found that the stability and accuracy of the model are very high even when D is lower than what is required for the injectivity of β_A. From these experimental observations, we deduce that there is still a scope further to investigate the complexity, necessary for successful sorting-based schemes.
- We noticed that increasing the embedding dimension D consistently improves both the accuracy and the stability of the model. The dependence of stability and accuracy on n and d are less clear and further work needs to be done to analyze their relationship.

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